NORTHWIND TRADERS | GLOBAL SALES ANALYSIS

CIS 235 | Team 3A | Spring 2024

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Executive Summary:

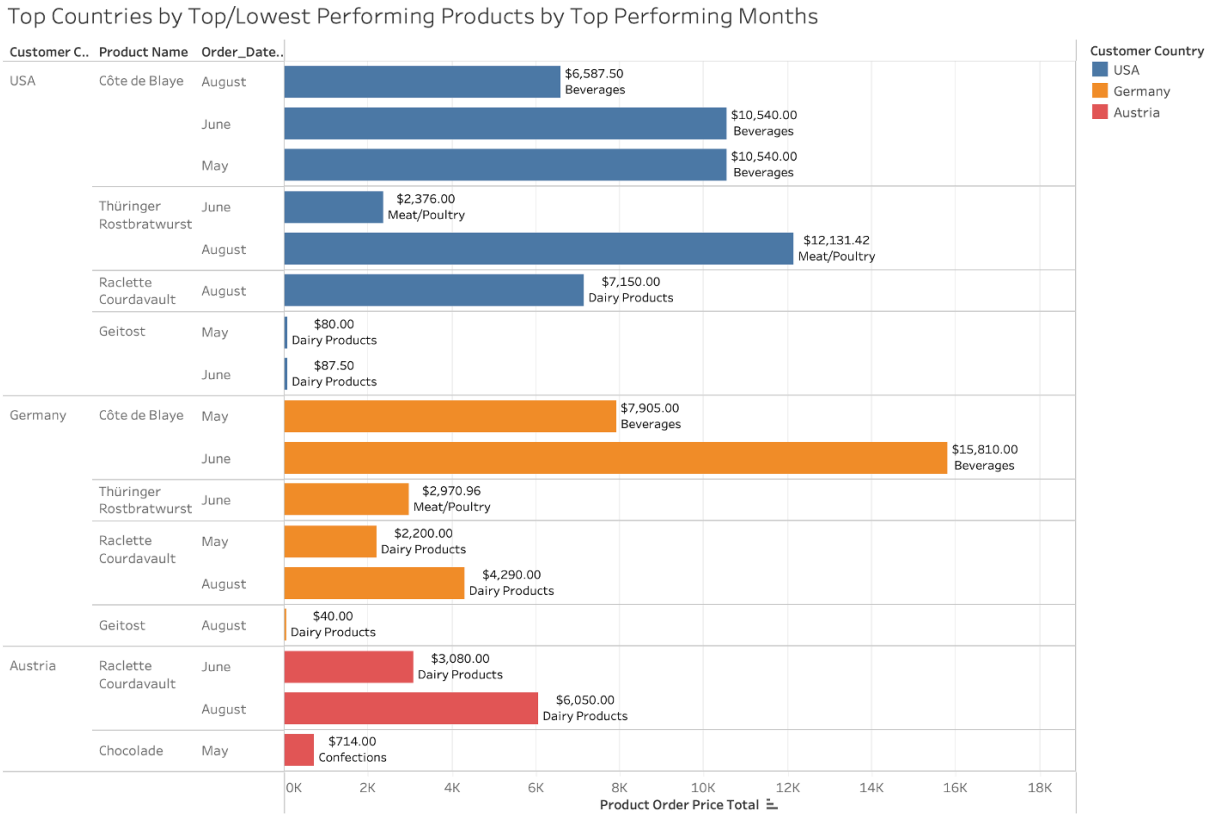


Team 3A Video Presentation

Northwind Trader Global Sales Data 20.21.23

- Top performing countries: USA ($926,144), Germany ($810,632), Austria ($492,699)
- Top performing product categories: Beverages ($287,607), Dairy Products ($251,331), Confections ($179,888)
- Top performing products: Côte de Blaye ($149,984), Thüringer Rostbratwurst ($87,736) Raclette Courdavault ($76,296)
- Top performing customers: QUICK-Stop ($117,483.39), Save-a-lot Markets($115,673.39), Ernst Handel($113,236.68)
- Top performing months: August(81.02% above avg.), May(49.09% above avg.), June(31.92% above avg.)
- Top 10 customers accounting for 46% of monetary value

Top Performing Products

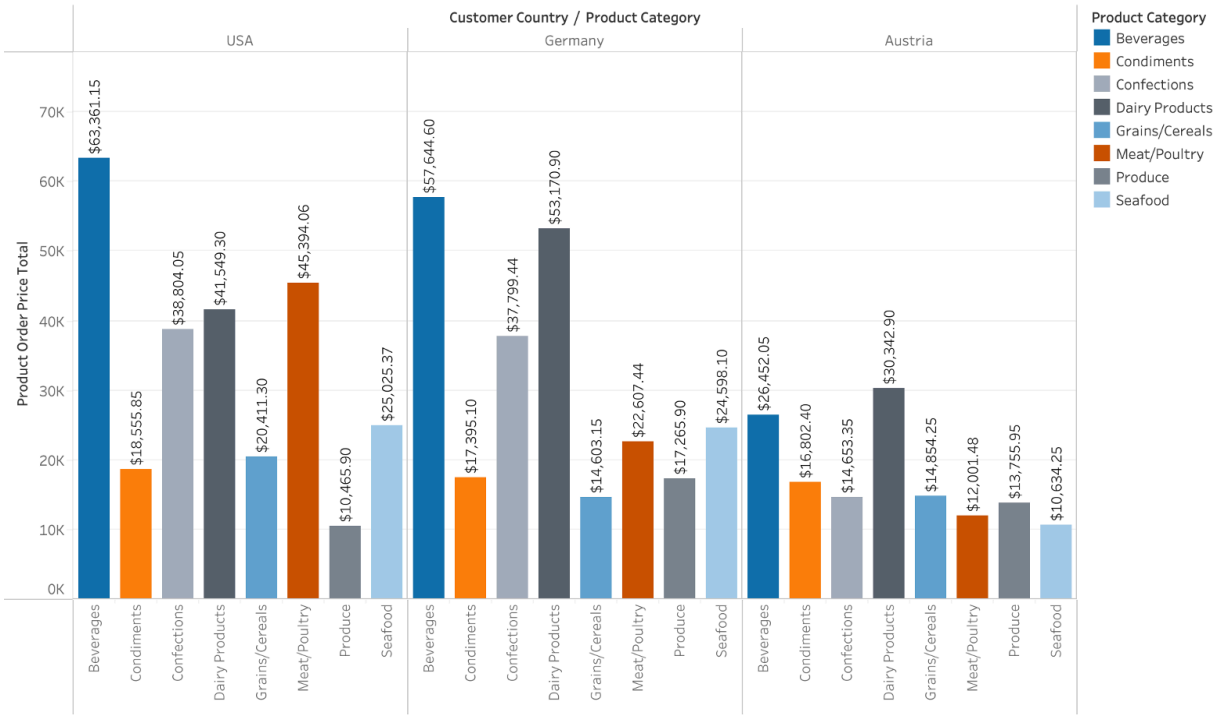


Sum of Product Order Price Total for each Order_Date_Month broken down by Customer Country and Product Name. Color shows details about Customer Country. The marks are labeled by sum of Product Order Price Total and Product Category. The view is filtered on Product Name, Customer Country and Order_Date_Month. The Product Name filter keeps 6 of 77 members. The Customer Country filter keeps Austria, Germany and USA. The Order_Date_Month filter keeps August, June and May.

What we know:

- **Top performing products**: Côte de Blaye ($149,984) Thüringer Rostbratwurst ($87,736) Raclette Courdavault ($76,296)
    - Côte de Blaye (May, June, August): **18.5%** sold in USA, **15.8%** Germany, **0%** Austria
    - Thüringer Rostbratwurst (May, June, August): **16.5%** sold in USA, **3.4%** Germany, **0%** sold in Austria
    - Raclette Courdavault (May, June, August): **9.4%** sold in USA, **8.5%** sold in Germany, **11.9%** sold in Austria
- **Lowest performing products** by product order price total **1**. Genen Shouyu ($1,814) **2**. Geitost ($1,714) **3**. Chocolade ($1,543)
    - Genen Shouyu (May, June, August): **0%** sold in USA, **0%** sold in Germany, **0%** sold in Austria
    - Geitost (May, June, August): **9.8%** sold in USA, **2.3%** sold in Germany, **0%** sold in Austria
    - Chocolade (May, June, August): **0%** sold in USA, **0%** sold in Germany, **46.3%** sold in Austria

Category Distribution by Top 3 Countries

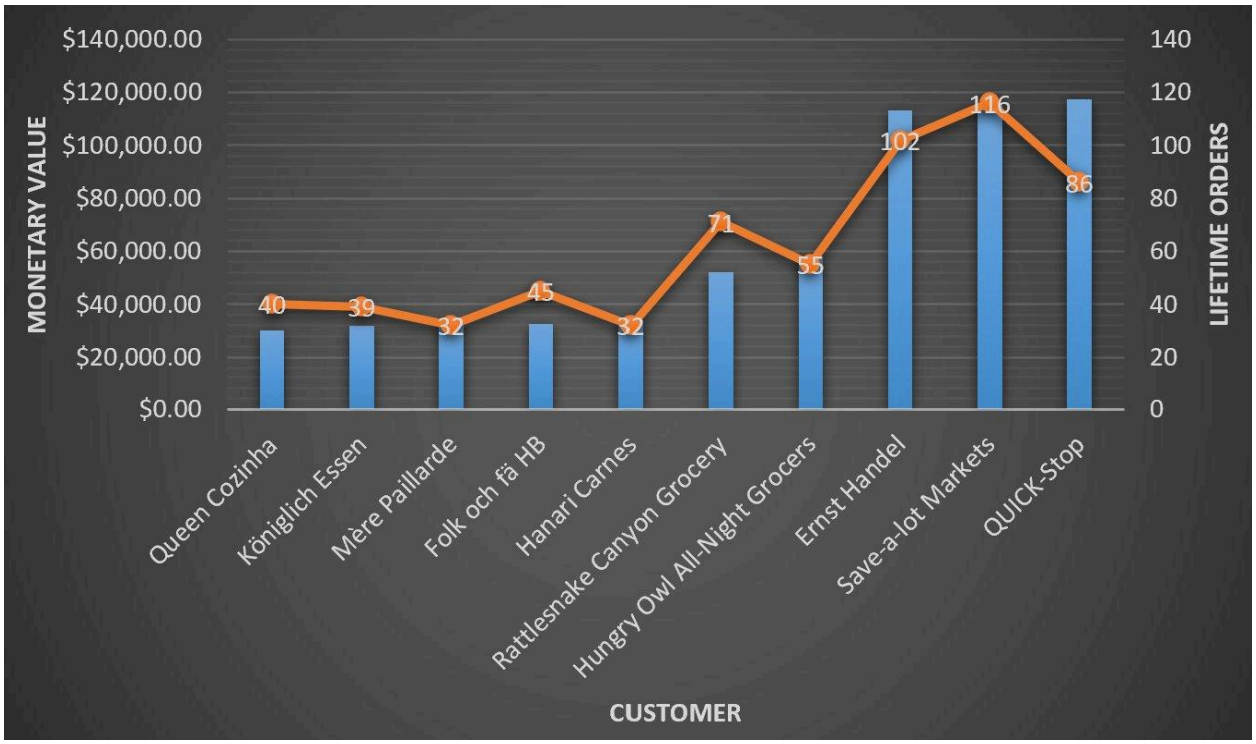


Sum of Product Order Price Total for each Product Category broken down by Customer Country. Color shows details about Product Category. The marks are labeled by sum of Product Order Price Total. The view is filtered on Customer Country, which keeps Austria, Germany and USA.

What we know:

- **Top Performing Products by Product Order Price Total**:
  - **Beverages** - USA ($63,361.15), Germany ($57,644.60) and Austria ($26,452.05).
  - **Confections**- USA($38,804.05), Germany($37,799.44), Austria ($14,653.35)
  - **Dairy Products** - USA($41,549.30),Germany($53,170.90), and Austria ($30,342.90)
- **Lowest Performing Products by Product Order Price Total**:
  - **USA**- Produce($10,465.90), Condiments($18,555.85), Grains/Cereals ($20,411.30)
  - **Germany**- Grains/Cereals ($14,603.15), Produce ($17,265.90), and Condiments ($17,395.10)
  - **Austria**- Seafood ($10,634.25), Meat/Poultry ($12,001.48), and Produce ($13,755.95)

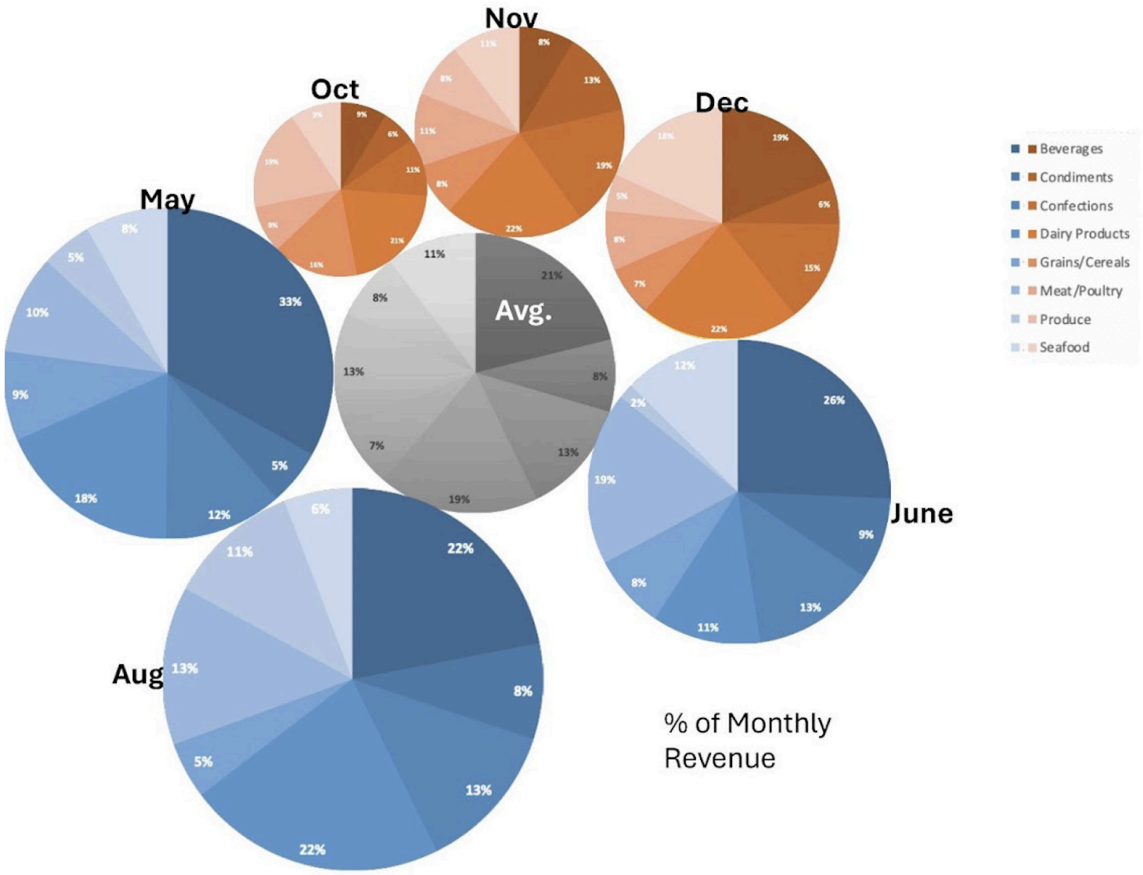
Customer, Frequency, Monetary Value (RFM) Analysis



What we know:

- RFM Analysis was performed to measure Recency, Frequency, and Monetary Value of customers. This allows us to put a value system to customers in order to shift focuses if needed, our report found:
    - Top 10 customers Accounted for 46% of Monetary Value
    - 12 Customers hold over 40 Orders each, appx 53% of sales
    - 7 customers order an average of over $1000 per order, twice the average of $487.13
    - Customers who ordered within the last 30 days have a total of 82% of all Monetary Value, implying customer retention is positive
    - 5 Customers at Risk Of Churn (Haven't ordered in 6+ months), with a current 2.2% Churn Rate (2 lost customers), We risk pushing this to 7.8%
    - Our 9th Biggest Customer “Mere Paillarde” is at risk of leaving after not placing an order for over 6 months
    - 53.9% of our customers fall under $10,000 MV despite an average of $15270.54, This indicates a huge spread in our customer base including super high outlying customers.

Product Category Distribution by Top/Bottom Months



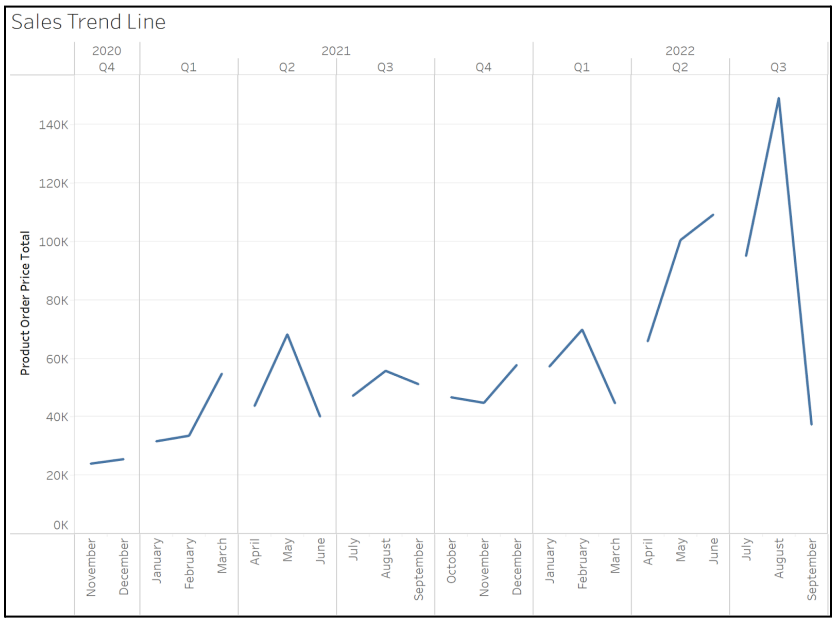
What we know:

- **Top performing months** are August(81.02% above avg.), May(49.09% above avg.), June(31.92% above avg.)
    - **Top performing categories** August and May: Beverages, Dairy June: Beverages, Meat/Poultry
    - **Lowest performing categories** August: Grains/Cereal, Seafood May: Produce, Condiments June: Produce, Grains/Cereal
- **Lowest performing months** are October(58.74% below avg.), November(39.21% below avg.), December(26.42% below avg.)
    - **Highest performing categories** October: Dairy, Produce November: Dairy, Confections December: Seafood, Beverages, Dairy
    - **Lowest Performing categories** October: Confections, Meat/Poultry November: Beverages, Grains/Cereal, Produce December: Grains/Cereal, Produce

What we need to know:

It would be helpful to know the cost of each product. With this we would be able to better analyze what categories have the best margins.

Sales Trend Line



What we know:

- The Sales Trend Line graph shows a sales trend for Northwind Trader from Q4 2020 to Q3 2022.
- There are sales peaks in the second and third quarters of 2022, indicating a strong seasonal effect on consumer buying patterns.
- The peak of sales occurs in August 2022, when the total product order price exceeds $140,000, indicating it's a top performing month
- Low-performing months are seen in the trend line, indicating lower sales volume and pointing to probable off-peak seasons for the organization

What we need to know:

- Insights into marketing and sales methods used during peak seasons may assist reproduction success in low-performing months.

Conclusion

1. Top Countries: United States, Germany, Austria
2. Top Customers: Ernst Handel, Save-a-lot Markets, and QUICK-Stop
3. Top Categories: Beverages, Dairy, Confections
4. Top Products:Côte de Blaye, Thüringer Rostbratwurst, Raclette Courdavault
5. Top Months: August, May, June
6. Bottom Countries: Poland, Norway, Argentina
7. Bottom Customers: Centro Comercial Moctezuma, Lazy K Kountry Store, Laughing Bacchus Wine Cellars
8. Bottom Categories: Grains/Cereal, Produce, Condiments
9. Bottom Products: Chocolade, Geitost, Genen Shouyu
10. Bottom Months: October, November, December

Moving forward Team 3A recommends an RFP to focus on strategic investments to analyze product, country, order, and price trends. By leveraging data analytics, they can gain insights into customer preferences, behaviors, and associated marginal costs, enabling targeted marketing efforts and product optimization.

Company Contacts

1. **Oracle Cloud Data**
    a. Contact Type: Audit
    b. External Contact: US West (Phoenix), AZ
        i. Website: https://www.oracle.com
        ii. Contact Info (623-555-5555)
2. **Pitney Bowes**
    a. Contact Type: Logistics
    b. External Contact: Stamford, CT
        i. Website:https://www.pitneybowes.com
        ii. Contact Info (203-777-7777)
3. **Neilson**
    a. Contact Type: Retail Audit
    b. External Contact: Chicago, IL
        i. Website:https://www.nielsen.com
        ii. Contact Info: (224-333-3333)

Name: Team 3A, Guggemos, MW 10:10-11	Proposal Due Date: 4/5/2024

Project Overview:

Team 3A seeks auditors to gather data spanning from the last recorded transaction in September 2022 to present day in April 2024 on supply chain expenses and efficiency including unit prices, bulk discounts, and timeliness for order processing, shipping, and delivery. Over this eighteen-month period, Team 3A aims to pinpoint any fluctuations in marginal costs within the current supply chain, as well as identify patterns in order processing and shipping to optimize operational costs. Bidding occurs between Oracle Cloud Data, Neilson, and Pitney Bowes.

Project Goals:

The primary objective of this Request for Proposal (RFP) is to acquire comprehensive information about our products and their associated costs from potential suppliers/vendors. By obtaining this data, we aim to gain insights into the marginal aspects of our project, enabling informed decision-making and strategic planning.

Key Deliverables:

1. Transparent cost structures for each product, including unit prices, bulk discounts, and any additional fees.
2. Documentation on quality control measures and compliance standards adhered to by the supplier.
3. Timelines for order processing, shipping, and delivery to ensure alignment with our project deadlines.

Expected Outcomes:

1. Enhanced understanding of the market landscape and competitive pricing structures.
2. Identification of cost-saving opportunities and potential areas for negotiation with suppliers.
3. Optimization of procurement processes to streamline operations and improve efficiency.
4. Improved ability to forecast project expenses and allocate resources effectively.
5. Strengthened relationships with suppliers through transparent communication and collaboration.

Relevance: Detailed product and cost information are crucial for data-driven decision-making and project optimization. By maximizing profits and cutting costs through strategic procurement, we aim to bolster our competitive position and drive sustainable growth.

Current Roadblocks and Barrier to Success:

1. **Market and Product Performance insight:**
   Inadequate insight into real-time market trends and product performance across locations and client segments, preventing proactive strategy modifications.
2. **Limited Access to Competitive Benchmarking Information:**
   Getting detailed information on how our products and services stack up against those of our rivals can be difficult.
3. **Limitations on Resources and Cost and Quality control:**
   Encountering restrictions on the amount of money, labor, and technology available for effective execution and growth. Cost and quality control issues include unclear cost structures for products and services, as well as uneven quality control and compliance documents.
4. **Customer Segmentation and Retention Value Analysis:**
   Difficulties in properly segmenting the customer base and implementing targeted retention tactics, particularly among high-value customers at risk of turnover. And inability to accurately assess the profitability and actual worth of customer connections, influences the ability to make strategic decisions.
5. **Seasonality and Demand Forecasting Issues:**
   Difficulty anticipating and planning for seasonal increases in demand across many areas, which affects inventory management and system capacity.
6. **Inefficiencies in Supply Chain and Distribution:**
   Difficulties optimizing order size, lowering transportation costs, and enhancing the overall efficiency of our supply and value chains.

Evaluation Metrics and Criteria:

Cost Transparency:

1. Clarity of information provided regarding unit prices for each product.
2. Transparency in bulk discounts offered.
3. Identification and disclosure of all additional fees associated with products or services.

Quality Control and Compliance Documentation:

1. Adequacy and comprehensiveness of quality control measures outlined.
2. Documentation demonstrating adherence to relevant compliance standards.
3. Transparency in communicating any deviations from industry norms or regulations.

Timeliness and Alignment with Project Deadlines:

1. Accuracy of timelines provided for order processing, shipping, and delivery.
2. Consistency between promised timelines and actual performance.
3. Ability to accommodate and meet project deadlines effectively.

Accuracy and Reliability of Data:

1. Precision and correctness of information provided.
2. Consistency and coherence of data across all provided documents.
3. Identification and rectification of any discrepancies or errors in the data.

Communication and Responsiveness:

1. Responsiveness of the supplier to queries and requests for clarification.
2. Clarity and effectiveness of communication channels utilized.
3. Proactive communication regarding any potential issues or delays.

Comprehensiveness of Information:

1. Inclusion of all necessary details relevant to the evaluation criteria.
2. Clarity and completeness of documentation provided.
3. Accessibility and ease of understanding of the information presented.

Project Timeline Detail selection requirements:

Transparency of Cost Structures:

1. Clear breakdown of unit prices for each product offered. Bulk discount rates, if applicable.
2. Disclosure of any additional fees or charges.

Product and Cost Information:

1. Comprehensive analysis of the market landscape and competitive pricing structures.
2. Identification of potential cost-saving opportunities and areas for negotiation with suppliers.

Documentation on Quality Control Measures and Compliance Standards:

1. Detailed documentation on quality control procedures implemented by the supplier.
2. Evidence of adherence to relevant compliance standards and regulations.

Timelines for Order Processing, Shipping, and Delivery:

1. Clear timelines provided for order processing from placement to fulfillment.
2. Estimated shipping and delivery times to ensure alignment with project deadlines.

Trend Identification and forecasting:

1. Utilize historical data and market research to identify patterns and trends
2. Employ statistical models and forecasting techniques such as time series and regression analysis.
3. Forecasting and trend identification for unobtained data 9/22 - 4/24

Project Timeline:

Phase 1 (Payout: 10% upon completion)

April 2024

1. Begin Data Analysis: The last recorded transaction marks the beginning of the data collection period for supply chain expenses and efficiency analysis.
2. Data Gathering Phase 1: Initial collection of data on unit prices, bulk discounts, and timeliness for order processing, shipping, and delivery from Oracle Cloud Data, Neilson, and Pitney Bowes.

May 2024

1. Interim Analysis: Perform a preliminary analysis to identify any fluctuations in marginal costs and patterns in order processing and shipping within the first six months of data collection.
2. Data Gathering Phase 2: Continue the collection of detailed data on supply chain expenses and efficiency.

Phase 2 (Payout: 30% upon completion)

June 2024 - July 2024

1. Secondary Analysis: Conduct a comprehensive analysis of the accumulated data to pinpoint cost-saving opportunities and areas for negotiation with suppliers.
2. Strategic Planning Meeting: Meet with stakeholders to review findings, discuss potential strategies for cost reduction, and initiate procurement process optimization.

Phase 3 (Payout: 60% upon completion)

August 2024

1. Vendor Negotiations: Engage in discussions with Oracle Cloud Data, Neilson, and Pitney Bowes to negotiate terms based on insights gained from the data analysis.

September 2024

1. Finalization of Vendor Contracts: Complete negotiations and finalize agreements with selected suppliers/vendors.
2. Project Completion: Submit a comprehensive report detailing the outcomes of the procurement process, including cost savings achieved, efficiency improvements, and strengthened supplier relationships.

	Budget Amount: $950,000